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Batch : T12
Subject : ASD&D

EXPERIMENT NO. 07

Aim: To implement monitoring using Prometheus / Grafana / Nagios

LO Mapping: LO5

LO5: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

Theory:

Continuous monitoring is a process to detect, report, respond all the attacks which occur in its infrastructure. Once the application is deployed into the server, the role of continuous monitoring comes in to play. The entire process is all about taking care of the company's infrastructure and respond appropriately.

Why We Need Nagios tool?

Here, are the important reasons to use Nagios monitoring tool:

- Detects all types of network or server issues
- Helps you to find the root cause of the problem which allows you to get the permanent solution to the problem
- Active monitoring of your entire infrastructure and business processes
- Allows you to monitor and troubleshoot server performance issues
- Helps you to plan for infrastructure upgrades before outdated systems create failures
- You can maintain the security and availability of the service
- Automatically fix problems in a panic situation
- Nagios is the most popular, open source, powerful monitoring system for any kind of infrastructure. It enables organizations to identify and resolve IT infrastructure problems before they affect critical business processes. Nagios has the capability of monitoring applications, services and entire IT infrastructure.
- NRPE is known as Nagios Remote Plugin Executor. It is designed to execute plugins on remote systems. On Windows hosts, the equivalent remote agent is NSClient++, which listens for Nagios requests and executes local checks (disk, CPU, memory, services, etc.).

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- NSClient++ runs as a Windows service on the remote host and waits for requests from the Nagios server (usually over the NRPE protocol, which NSClient++ also supports). When the Nagios server needs to check a resource on that host, it sends a command; NSClient++ executes the check locally and returns the result to Nagios.

Note on running Nagios on Windows

Nagios Core is built for Linux and does not install natively on Windows. The standard practice for a Windows machine is to run the Nagios Core server inside a Docker container (using Docker Desktop with the WSL2 backend), and to monitor Windows hosts using NSClient++, which is the Windows counterpart of NRPE. All commands below are Windows PowerShell / Command Prompt commands.

1 - Pre-requisite: Install Docker Desktop

Download and install Docker Desktop for Windows and enable the WSL2 based engine. Verify the installation from PowerShell:

```
PS C:\Users\vishal> docker --version
PS C:\Users\vishal> docker version
```

Also make sure the WSL2 feature is enabled:

```
PS C:\Users\vishal> wsl --install
PS C:\Users\vishal> wsl --set-default-version 2
```

2 - Pull and Run the Nagios Core Docker Image

Pull a ready-made Nagios Core image and run it as a container, mapping the web port to the host.

```
PS C:\Users\vishal> docker pull jasonrivers/nagios:latest
PS C:\Users\vishal> docker run --name nagios4 -d -p 8080:80 jasonrivers/nagios:latest
```

Verify the container is up and running:

```
PS C:\Users\vishal> docker ps
```

3 - Set the Apache/Nagios Admin Password

Set (or reset) the password for the nagiosadmin user inside the running container:

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```
PS C:\Users\vishal> docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin
```

Enter and confirm the password when prompted (e.g. 123456).

4 - Access Nagios Web Interface

Access the Nagios web interface from a browser on the Windows host:

```
http://localhost:8080/nagios/
```

Prompting for Authentication Password -

username: nagiosadmin

Password: 123456 (which you entered while configuration)

5 - Install NSClient++ on the Windows Remote Host

On the Windows machine you want to monitor, download and silently install NSClient++ (the Windows equivalent of the NRPE agent):

```
PS C:\Users\vishal> Invoke-WebRequest -Uri "https://github.com/mickem/nscp/releases/latest/download/NSCP-0.5.2-x64.msi" -OutFile "NSCP.msi"
PS C:\Users\vishal> Start-Process msiexec.exe -ArgumentList '/i NSCP.msi ADDLOCAL=ALL MODULES=CheckExternalScripts,NRPEServer /qn' -Wait
```

6 - Configure NSClient++

Edit the NSClient++ configuration file and add the Nagios server's IP address to the allowed hosts list:

```
PS C:\Program Files\NSClient++> notepad nsclient.ini
allowed hosts=127.0.0.1,192.168.1.100
```

Where 192.168.1.100 is your Nagios server (Docker host) IP address.

After saving the file, restart the NSClient++ service so the change takes effect:

```
PS C:\Users\vishal> Restart-Service nscp
```

7 - Verify Connectivity from Nagios

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From the Nagios server (inside the Docker container), run the NRPE check against the Windows remote host to confirm connectivity. Here 192.168.1.50 is the remote Windows host's IP address.

```
PS C:\Users\vishal> docker exec -it nagios4 /opt/nagios/libexec/check_nrpe -H 192.168.1.50
NRPE v2.15
```

8 - Add the Windows Host in Nagios

Create a configuration file for the new host inside the Nagios container. We also define a service that continuously checks whether the host is reachable (PING check).

```
PS C:\Users\vishal> docker exec -it nagios4 sh -c "vi /opt/nagios/etc/servers/MyWindowsHost001.cfg"
define host {
    use             windows-server
    host_name       Windows_Host_001
    alias           Windows Host 001
    address         192.168.1.50
    register        1
}
define service{
    host_name               Windows_Host_001
    service_description     PING
    check_command           check_ping!100.0,20%!500.0,60%
    max_check_attempts      2
    check_interval          2
    retry_interval          2
    check_period            24x7
    check_freshness         1
    contact_groups          admins
    notification_interval   2
    notification_period     24x7
    notifications_enabled   1
    register                1
}
```

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Verify the configuration file for errors, then restart the Nagios container so the changes take effect:

```
PS C:\Users\vishal> docker exec -it nagios4 /opt/nagios/bin/nagios -v /opt/nagios/etc/nagios.cfg
PS C:\Users\vishal> docker restart nagios4
```

9 - Check Host in Nagios Web Interface

Open the Nagios web interface (http://localhost:8080/nagios/) and check that the new Windows host has been added and is being monitored under Current Status -> Hosts.

Output :

```
PS C:\Users\kunal> docker --version
Docker version 29.7.2, build a7dcaa6
PS C:\Users\kunal> docker version
Client:
 Version:           29.7.2
 API version:       1.55
 Go version:        go1.26.5
 Git commit:        a7dcaa6
 Built:             Wed Aug  5 18:31:33 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.90.0 (238679)
 Engine:
  Version:          29.7.2
  API version:      1.55 (minimum version 1.40)
  Go version:       go1.26.5
  Git commit:       6a43e3d
  Built:            Wed Aug  5 18:28:36 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.3.3
  GitCommit:        aad11006b869517fcd3009450b6f82da282e1a9b
 runc:
  Version:          1.4.3
  GitCommit:        v1.4.3-0-gbb14dabe
 docker-init:
  Version:          0.19.0
```

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```
PS C:\Users\kunal> wsl --install
A distribution with the supplied name already exists. Use --name to chose a different name.
Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS
PS C:\Users\kunal> wsl --set-default-version 2
For information on key differences with WSL 2 please visit https://aka.ms/wsl2
The operation completed successfully.
PS C:\Users\kunal> docker pull jasonrivers/nagios:latest
latest: Pulling from jasonrivers/nagios
6fe99ef1bed3: Pull complete
cb259a83ac3d: Pull complete
e68fcda12c0c: Pull complete
dc3a88012a3e: Pull complete
a6b90b5bbb15: Pull complete
c42073aaf047: Pull complete
f47c3af6bf44: Pull complete
54b9e993e315: Pull complete
1b9771f55a12: Pull complete
748b632e75e7: Pull complete
59215b8e47cf: Pull complete
d003d4303177: Pull complete
fdbaa15b8986: Pull complete
80b6ab7aef5d: Pull complete
a7bd6dce6a77: Pull complete
a80db686d641: Pull complete
f2f9a1b3ce48: Pull complete
7edde1931fd2: Pull complete
0d59a81a5608: Pull complete
37f75c8146e2: Pull complete
1413dee23aae: Pull complete
c5701b5e1540: Pull complete
e4dc1d4e010c: Pull complete
```

```
PS C:\Users\kunal> docker exec -it nagios4 /opt/nagios/libexec/check_nrpe -H 192.168.0.102 -2
I (0.19.0 2026-09-07) seem to be doing fine...

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug nagios4
    Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\kunal> docker exec -it nagios4 sh
# ls /opt/nagios/etc/servers
ls: cannot access '/opt/nagios/etc/servers': No such file or directory
# ls /opt/nagios/etc
cgi.cfg  conf.d  htpasswd.users  monitor  nagios.cfg  nsca.cfg  objects  resource.cfg  send_nsca.cfg
```

```
PS C:\Users\kunal> docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED         STATUS         PORTS
           NAMES
7e20536f4f8a   jasonrivers/nagios:latest   "/usr/local/bin/star…"   53 minutes ago   Up 14 minutes   0.0.0.0:8080->80/tcp, [::]:8
080->80/tcp   nagios4
```

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```
# cat /opt/nagios/etc/conf.d/windows.cfg
define host {
    use                     generic-host
    host_name               Windows_Host_001
    alias                   My Windows Computer
    address                 192.168.0.102
    max_check_attempts      3
    check_period            24x7
    notification_interval   30
    notification_period     24x7
}

define service {
    use                     generic-service
    host_name               Windows_Host_001
    service_description     PING
    check_command           check_ping!100.0,20%!500.0,60%
}
# grep -n "conf.d" /opt/nagios/etc/nagios.cfg
1347:cfg_dir=/opt/nagios/etc/conf.d
# /opt/nagios/bin/nagios -v /opt/nagios/etc/nagios.cfg

Nagios Core 4.5.12
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2026-03-25
License: GPL
```

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```
# /opt/nagios/bin/nagios -v /opt/nagios/etc/nagios.cfg

Nagios Core 4.5.12
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Website: https://www.nagios.org
Reading configuration data...
   Read main config file okay...
   Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
        Checked 8 services.
Warning: Host 'Windows_Host_001' has no default contacts or contactgroups defined!
        Checked 2 hosts.
        Checked 1 host groups.
        Checked 0 service groups.
        Checked 1 contacts.
        Checked 1 contact groups.
        Checked 23 commands.
        Checked 5 time periods.
        Checked 0 host escalations.
        Checked 0 service escalations.
Checking for circular paths...
```

```
Checking for circular paths...
        Checked 2 hosts
        Checked 0 service dependencies
        Checked 0 host dependencies
        Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 1
Total Errors:   0

Things look okay - No serious problems were detected during the pre-flight check
# exit

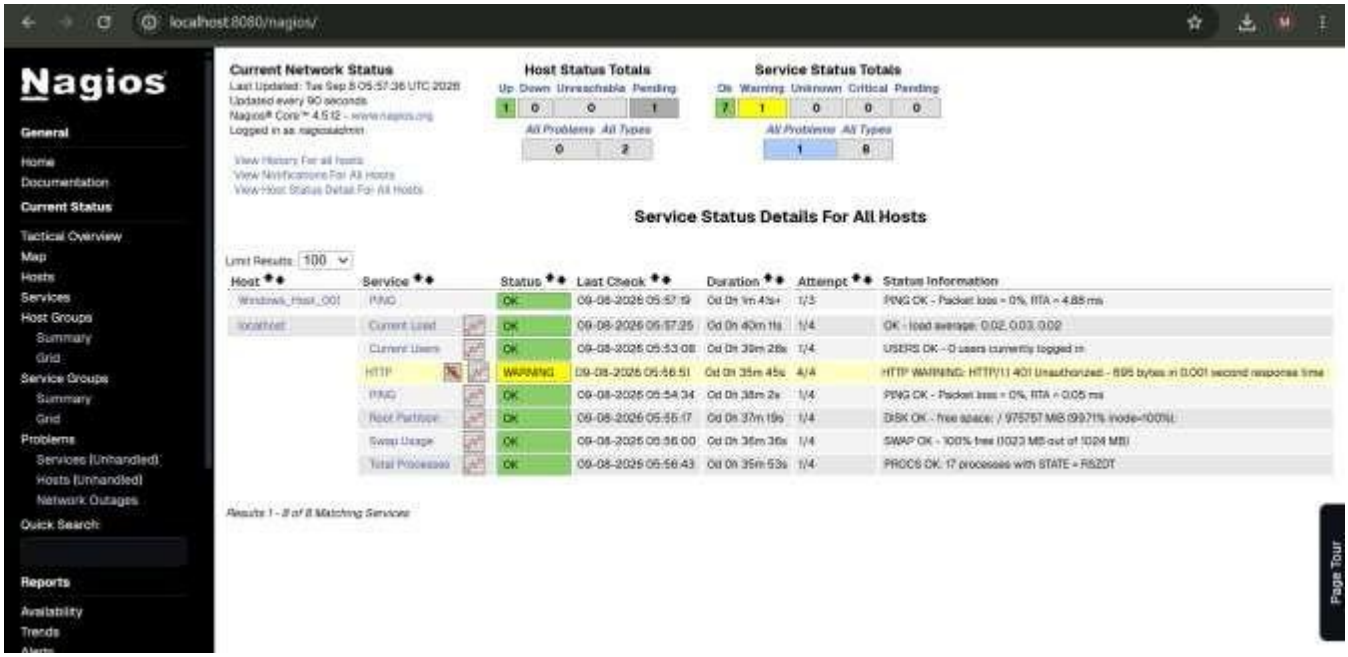
What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug nagios4
    Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\kunal> docker restart nagios4
nagios4
```

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Conclusion:

The experiment successfully demonstrated continuous infrastructure monitoring using **Nagios**. Nagios was deployed inside a Docker container, while **NSClient++** was used to enable monitoring of the Windows system. The successful display of the Windows host and PING service confirmed that the monitoring configuration was working correctly. This experiment shows how monitoring tools help detect system and network problems and maintain the availability and reliability of IT infrastructure.